

CHECK EXISTING RAFTERS FOR ADDED LOAD

ASSUME MAX DL ONLY ON HANGER + ROOF TL W

$$RF - LL - W_{LL} = 2.0' \times 16 \frac{\#}{10'} = 32.0 \frac{\#}{10'} \\ (10 IN 12 SLOPE \pm)$$

$$W_{DL} = 28.0 \frac{\#}{10'}$$

$$W_{LL} = 32.0$$

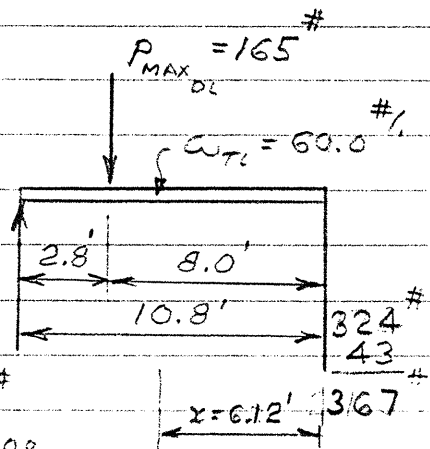
$$W_{TL} = 60.0 \frac{\#}{10'}$$

$$+M_{MAX} \text{ AT } X = 367/60 = 6.12'$$

$$+M_{MAX} = 367 \times 6.12 - 60 \times 6.12^2/2$$

$$= 1122 \frac{\#}{10'}$$

$$W_{TL} \begin{array}{r} 324 \\ P_{MAX} \text{ DL } 122 \\ \hline 446 \end{array} \frac{\#}{10'}$$



$$+M_{MAX} = 446 \times 4.68 - 60 \times 4.68^2/2 - 165 \times 1.88 \\ = 1075 \frac{\#}{10'}$$

$$f_b = 12 \times 1122 / 12.6 = 1069 \frac{\#}{10'} \text{ OK}$$

$$SHEAR: f_v = 1.5 \times 485 / 13.8 = 53 \frac{\#}{10'} \text{ OK}$$

LOAD TO BOLTS AT 2 X 6 LEDGER

$$W_{TL} = 1.5' \times (5.0 \frac{DL}{10'} + 2.0 \frac{FL}{10'} + 4.0 \frac{LED}{10'} + 4.0 \frac{LL}{10'}) \times 12.2 = 260 \frac{\#}{10'}$$

$$P_{92W} = 2 \times 40 \frac{\#}{10'} = 80 \frac{\#}{10'}$$

$$P_{38W} = 6 \times 10 \frac{\#}{10'} = 60 \frac{\#}{10'}$$

$$P_E = 2 \times 15 \frac{\#}{10'} = 30 \frac{\#}{10'}$$

$$TL = 1030 \frac{\#}{10'}$$

FOR 5 BOLTS

$$BOLT \text{ LOAD} = 206 \frac{\#}{10'} \rightarrow \text{OK}$$